

Problem 7

Where do the points lie in the complex plane for which the following conditions are satisfied:

$$\begin{aligned} 1^\circ \quad & 2 < |z| < 3, \quad \frac{\pi}{4} < \varphi < \frac{\pi}{3}; \quad 2^\circ \quad |z| > 1, \quad \frac{\pi}{3} < \varphi < \frac{\pi}{2}; \\ 3^\circ \quad & |z - z_0| < 1; \quad 4^\circ \quad |z - i| < 1 \text{ and } |z - 1| < 1; \quad 5^\circ \quad |z| + \operatorname{Re}(z) < 2. \end{aligned}$$

Solution

Part 1°: $2 < |z| < 3, \quad \frac{\pi}{4} < \varphi < \frac{\pi}{3}$

Analysis

This condition consists of two parts:

- $2 < |z| < 3$: The modulus is between 2 and 3
- $\frac{\pi}{4} < \varphi < \frac{\pi}{3}$: The argument is between $\frac{\pi}{4}$ (45°) and $\frac{\pi}{3}$ (60°)

Geometric Description

The region is an **annular sector** (a "slice" of an annulus):

- Between two concentric circles: radius 2 (inner) and radius 3 (outer)
- Within the angular sector from $\frac{\pi}{4}$ to $\frac{\pi}{3}$
- Located in Quadrant I

Answer

Annular sector: $2 < |z| < 3$ with argument between 45 and 60

The region is bounded by:

- Inner circle: $|z| = 2$
- Outer circle: $|z| = 3$
- Ray at angle $\frac{\pi}{4}$
- Ray at angle $\frac{\pi}{3}$

Part 2°: $|z| > 1, \quad \frac{\pi}{3} < \varphi < \frac{\pi}{2}$

Analysis

- $|z| > 1$: Points outside the unit circle
- $\frac{\pi}{3} < \varphi < \frac{\pi}{2}$: Argument between $\frac{\pi}{3}$ (60°) and $\frac{\pi}{2}$ (90°)

Geometric Description

An **unbounded angular sector**:

- Exterior of the unit circle (radius > 1)
- Within the angular sector from 60 to 90
- In Quadrant I, closer to the positive imaginary axis

Answer

Unbounded sector: $|z| > 1$ with argument between 60 and 90

The region extends to infinity in the angular sector, bounded by:

- Inner boundary: unit circle $|z| = 1$
- Ray at angle $\frac{\pi}{3}$
- Ray at angle $\frac{\pi}{2}$ (positive imaginary axis)

Part 3°: $|z - z_0| < 1$

Analysis

The condition $|z - z_0| < 1$ means:

distance from z to z_0 is less than 1

Geometric Description

This is the **interior of a circle**:

- Center at point z_0 in the complex plane
- Radius = 1
- Does not include the boundary circle itself

Answer

Open disk of radius 1 centered at z_0

If $z_0 = x_0 + iy_0$, then the region is:

$$\{(x, y) : (x - x_0)^2 + (y - y_0)^2 < 1\}$$

Part 4°: $|z - i| < 1$ and $|z - 1| < 1$

Analysis

Two simultaneous conditions:

- $|z - i| < 1$: Interior of circle centered at $i = (0, 1)$ with radius 1
- $|z - 1| < 1$: Interior of circle centered at $1 = (1, 0)$ with radius 1

We need the **intersection** of these two regions.

Geometric Description

The region is the **lens-shaped intersection** of two circles:

- Circle 1: center $(0, 1)$, radius 1
- Circle 2: center $(1, 0)$, radius 1

The two circles intersect at two points. We can find these intersection points:

For $z = x + iy$:

$$|z - i|^2 < 1 \implies x^2 + (y - 1)^2 < 1$$

$$|z - 1|^2 < 1 \implies (x - 1)^2 + y^2 < 1$$

The circles intersect when both equalities hold. The distance between centers is:

$$|i - 1| = |1 + i| = \sqrt{1 + 1} = \sqrt{2} \approx 1.414$$

Since $\sqrt{2} < 2$ (sum of radii), the circles do intersect.

Answer

Lens-shaped region: intersection of disks centered at i and 1 , each with radius 1

Part 5°: $|z| + \operatorname{Re}(z) < 2$

Analysis

Let $z = x + iy$. Then:

$$\begin{aligned}|z| + \operatorname{Re}(z) &< 2 \\ \sqrt{x^2 + y^2} + x &< 2 \\ \sqrt{x^2 + y^2} &< 2 - x\end{aligned}$$

For this to be valid, we need $2 - x > 0$, i.e., $x < 2$.

Squaring both sides (valid when $2 - x > 0$):

$$\begin{aligned}x^2 + y^2 &< (2 - x)^2 \\ x^2 + y^2 &< 4 - 4x + x^2 \\ y^2 &< 4 - 4x \\ y^2 &< 4(1 - x)\end{aligned}$$

For $x < 1$: $y^2 < 4(1 - x)$, which gives $|y| < 2\sqrt{1 - x}$

Geometric Description

Rearranging:

$$\begin{aligned}y^2 < 4 - 4x &\implies y^2 + 4x < 4 \\ y^2 = -4x + 4 &\implies y^2 = -4(x - 1)\end{aligned}$$

This is a **parabola** opening to the left with vertex at $(1, 0)$.

The region is the interior of this parabola (to the left of the vertex).

Answer

Interior of parabola: $y^2 < 4(1 - x)$ or $y^2 + 4x < 4$

The boundary is the parabola $y^2 = 4(1 - x)$ with vertex at $(1, 0)$.

Final Answer

- 1° Annular sector between circles $|z| = 2$ and $|z| = 3$, angles 45-60
- 2° Unbounded sector outside $|z| = 1$, angles 60-90
- 3° Open disk of radius 1 centered at z_0
- 4° Lens-shaped intersection of two unit disks at i and 1
- 5° Interior of parabola $y^2 = 4(1 - x)$ opening left from $(1, 0)$